Team Number:	apmcm25304033
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2025 APMCM summary sheet

This study delves into global trade disruptions and industrial restructuring from the U.S. ”Reciprocal Tariffs” policy since the 2020s. Centering on five core questions, it uses panel data econometrics, Logit models, dynamic simulation, and game theory, integrating 2017–2025 data to quantify impacts and predict trends.

Addressing Question 1, a two-stage framework is built: a log-log panel model estimates soybean import demand elasticity -1.02, adjusted for shocks like African Swine Fever, plus a tariff gradient simulator. Results show U.S. tariff hikes reduce its soybean exports; Brazil absorbs most diverted demand, Argentina’s gain limited by logistics and capacity.

Turning to Question 2, a multi-dimensional framework supply-side cost minimization plus demand-side Logit model includes Mexican transshipment costs, U.S. local production costs, and demand elasticity -3.0. Simulations: low Japanese car tariffs mean Mexican transshipment; high tariffs shift to U.S. production, raising car prices, shrinking demand, and risking jobs.

For Question 3, a Security-Efficiency model balances supply chain autonomy and inflation control, dividing chips into high-end, mid-end, and low-end categories. Testing Trump high tariffs without subsidies versus Biden moderate tariffs with high subsidies: Trump’s policy lowers high-end chip security and raises low-end inflation; Biden’s improves security and controls mid-end inflation, proving subsidies outperform tariffs for tech industries.

Shifting to Question 4, a dynamic difference-in-differences model short-term and long-term import elasticities predicts 2025–2028 tariff revenue: 2025 peak with net growth, then shrinking tax base; cumulative gains are large, long-term growth constrained by falling imports.

With Question 5, a multiple linear regression model quantifies retaliatory measures China’s rare earth and lithium controls, agricultural tariffs, using reshoring investment, material costs, and agricultural exports. Rising costs and falling exports offset tariffs’ reshoring effects, making the policy ineffective short-to-medium term.

Finally, the model is validated with historical data and robustness tests. This study aids tariff strategies and multilateral trade maintenance. A limitation: no sudden geopolitical shocks are included.

Keywords: U.S. Tariff Policies Global Trade Regional Economies Mathematical Modeling Trade Impact Analysis

Contents

I. Introduction

1.1 Background

Since the 2020s, driven by the "America First" trade ideology, U.S. tariff policy has undergone major adjustments. On April 2, 2025, the Trump administration introduced the "Reciprocal Tariffs" policy, establishing a 10% baseline tariff on all trading partners and higher tariffs on around 60 countries with which the U.S. ran trade deficits, including China. The stated goal was to reduce trade deficits and promote manufacturing reshoring, but the policy violated WTO principles, including Most-Favored-Nation (MFN) treatment and tariff preferences for developing countries.

The U.S. trade-weighted average tariff rose sharply from 2.44% at the start of 2025 to 20.11% by August 2025, reaching the highest effective level since 1933 (GACC[1]) . While tariffs can protect domestic industries and generate revenue, excessively high tariffs harm exports and raise domestic consumer prices. For example, after the policy's implementation, U.S. imports from China fell 35% and exports to China declined 18% year-on-year by May 2025, narrowing the trade deficit from $30.8 billion to $18.0 billion but negatively impacting U.S. industries.

Overall, the "Reciprocal Tariffs" policy reflects a shift toward protectionism, disrupts global trade rules, and may hinder long-term global economic growth. This study aims to evaluate the policy's effects on international trade, industrial development, and consumer welfare, providing a framework to assess unilateral tariff measures and inform strategic responses to preserve the multilateral trading system.

1.2 Question Restatement

By utilizing the attached U.S. foreign trade data (2020 onwards) and supplementary information collected from authoritative channels such as WTO reports and national customs statistics, we will conduct an in-depth analysis of how U.S. tariff adjustments reshape cross-border trade patterns and affect the economic operations of key regions. Combined with the current global trade order and the policy responses of major economies, we will put forward practical countermeasures and development suggestions for industries, such as soybean, automobile, and semiconductor, impacted by tariff changes.

1.2.1 *Question 1*

The U.S., Brazil and Argentina are top soybean exporters, and China the largest importer. Analyze their current soybean trade, build a model to assess U.S. tariff impacts on their soybean industries, and estimate post-adjustment export volume and value distribution.

1.2.2 *Question 2*

The U.S. is the second-largest auto market and top importer; 46% of its 2024 car sales were imports, mainly from Japan. Japan also produces in the U.S. or Mexico for U.S. export. Analyze

[1] [GACC, 2025] General Administration of Customs of the People's Republic of China, UN Comtrade, and USDA FAS (October 2025)

Japan's U.S. auto position, build a model with Japan's non-tariff responses to study U.S. tariff impacts on their auto trade, U.S. import structure and U.S. auto industry.

1.2.3 *Question 3*

The U.S. leads the global semiconductor industry but lacks strong manufacturing. Biden's 2022 CHIPS Act uses subsidies; Trump's 2025 admin claims tariffs work better. The U.S. also controls high-end chip exports to China. Build a model to analyze U.S. tariff effects on domestic semiconductor manufacturing and high/mid/low-end chip trade, considering efficiency and national security.

1.2.4 *Question 4*

Higher tariffs may raise U.S. short-term tariff revenue but cut trade volumes, lowering revenue long-term. Build a model to analyze short/medium-term tariff impacts on U.S. revenue and predict net changes in Trump's second term.

1.2.5 *Question 5*

U.S. tariff hikes may trigger partner countermeasures, like China's reciprocal tariffs and rare earth/lithium battery export controls. These affect U.S. trade, agriculture, industry and financial markets. Select/construct indicators to model tariff impacts on the U.S. economy and evaluate if "reciprocal tariffs" drive manufacturing reshoring.

II. Problem Analysis

2.1 Analysis of Question 1: Soybean Trade Dynamics

We evaluate the effects of U.S. tariff adjustments using a **Partial Equilibrium Framework**. Historical trade data (2017–2025) are filtered to identify structural breaks, after which a **Log-Log Demand Model** is estimated to obtain a stable price elasticity ($\eta \approx -1$).[2] A **Tariff Gradient Simulation** then quantifies the resulting trade diversion toward Brazil and Argentina under different tariff scenarios.

2.2 Analysis of Question 2: Automobile Supply Chain Reconfiguration

To assess whether tariffs induce reshoring or merely redirect trade through Mexico, we integrate a **Cost Minimization Model** with a **Multinomial Logit Demand System**. By decomposing manufacturing, logistics, and tariff components, the model estimates the tariff threshold at which firms shift from direct exports to transshipment and eventually to U.S. localization. This approach identifies the tipping point around which consumer prices and market shares begin to respond non-linearly.

2.3 Analysis of Question 3: Semiconductor Security–Efficiency Trade-off

We construct a **Security-Efficiency Equilibrium Model** to analyze the balance between supply-chain security and inflationary pressure across high-, mid-, and low-end chip tiers. Baseline costs are calibrated using a reverse-engineering method, and two policy mixes— high tariffs versus production

[2] Elasticity estimates follow standard agricultural trade models based on long-run log-log demand specifications.

subsidies—are simulated to evaluate how innovation capacity and market allocation change across tiers.

2.4 Analysis of Question 4: Dynamic Tariff Revenue Forecasting

To analyze fiscal outcomes under the "Reciprocal Tariffs," we employ a **Dynamic Difference-in-Differences** approach, comparing a CBO-style "Business-as-Usual" baseline with a tariff-shock scenario. A **Time-Decaying Elasticity** captures the transition from short-term rigidity to medium-run supply-chain adjustment, separating "Rate Effects" from long-run "Base Erosion."

2.5 Analysis of Question 5: Evaluation of Reshoring Effectiveness

To evaluate whether tariffs successfully promote reshoring under retaliation, we estimate a **Multivariate Correlation-Based Model**. Because available data are limited, a **Cholesky-based Synthetic Dataset** is generated to preserve historical covariance among key variables (e.g., critical mineral costs and export shocks).[3] The model assesses whether rising input costs from retaliation offset the protective benefits of tariffs and thus determines the net reshoring effectiveness.

III. Model Assumptions

1. Countries act as rational agents pursuing economic welfare or strategic objectives.
2. In the absence of major exogenous shocks, global supply–demand patterns remain stable over 2025–2029.
3. U.S. tariff policies under the "Reciprocal Tariffs" framework are fully enforced and remain unchanged unless specified; partner responses follow historical retaliation patterns with reasonable lags.
4. Exchange rates fluctuate within a moderate range; It is assumed that the Exchange Rate Pass-Through (ERPT) is incomplete, yet it follows the trend of relative purchasing power parity in the long-run equilibrium.
5. Products within relevant HS codes (soybeans, automobiles, semiconductors, etc.) are sufficiently homogeneous for aggregate elasticity estimation.
6. Domestic prices pass through tariff-inclusive import prices to varying degrees according to estimated elasticities.
7. Exporting countries' supply capacities adjust gradually from short-term rigidity through acreage, utilization, or investment.
8. Official statistical data are reliable; minor gaps are filled via credible interpolation, and ceteris paribus holds for variables not explicitly modeled.

[3] Cholesky-based Monte Carlo simulation is used purely for robustness testing, not causal inference.

IV. Description of the Symbols

Table 1 Main Symbols and Descriptions

Symbol	Description	Unit
General & Policy Variables		
τ	Effective tariff rate imposed on commodities	%
τ_{US}	U.S. "Reciprocal Tariffs" rate (Baseline + Additional)	%
R_{ex}	Exchange rate (e.g., JPY/USD, BRL/USD)	-
ε	Price elasticity of import demand / Dynamic elasticity	-
S_j	Market share of country j	%
Problem 1: Soybean Trade Model		
Q_{exp}	Export volume of soybeans	Million Tons
P_{cif}	CIF price adjusted for tariffs and exchange rates	USD/Ton
S^{prod}	Production capacity share of exporting countries	%
Problem 2 & 3: Industrial Supply Chain Model		
C_k	Landed cost or manufacturing cost for supply path k	USD/Unit
λ	Cost sensitivity coefficient in Logit choice model	-
F_{capex}	Amortized fixed costs for local production	USD
Z_{sec}	National Security Index (Semiconductor Model)	0–100 Score
I_{RnD}	Innovation Factor affected by export controls	0–1.0 Index
$S_{subsidy}$	Government subsidy amount (e.g., CHIPS Act)	USD/Unit
Problem 4 & 5: Macroeconomic & Revenue Model		
$V(t)$	Effective tax base (Total Import Value) at year t	Billion USD
R_{gov}	U.S. Federal tariff revenue	Billion USD
RI	Reshoring Investment Index	Index
$Cost_{input}$	Cost of critical inputs (Lithium, Rare Earths)	USD/Unit

Note: Minor auxiliary variables (e.g., regression coefficients β_i, time subscript t) are defined locally within the text.

V. Problem 1: Impact of U.S. Tariff Adjustments on Global Soybean Trade

5.1 Data Collection and Preprocessing

To quantitatively analyze the soybean trade dynamics between China and the three major exporters (United States, Brazil, and Argentina), we constructed a comprehensive dataset covering the period 2017–2025.

5.1.1 *Data Sources and Indicator Selection*

Through a literature review and data mining, we identified the key factors influencing export volumes. As illustrated in Figure 1, the data hierarchy focuses on the interplay between tariff policies, market prices, and supply capacities.

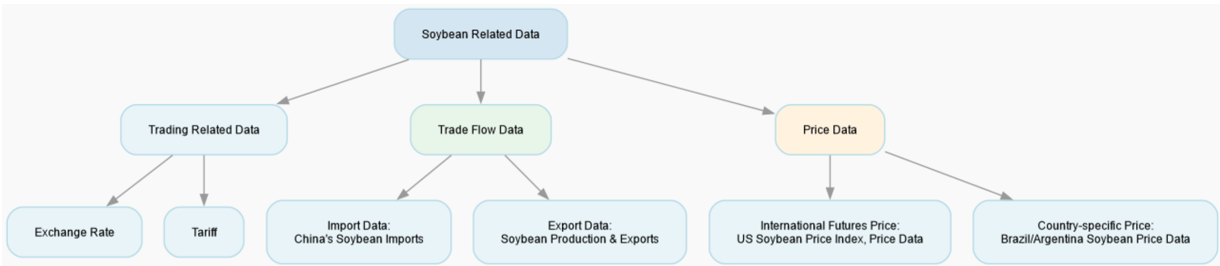


Figure 1 Data Hierarchy and Interaction Mechanism for Soybean Trade Analysis

Data was acquired from authoritative sources including the General Administration of Customs of the People's Republic of China (GACC), UN Comtrade, and the USDA Foreign Agricultural Service[cite: 58]. The dataset includes:

- **Trade Volume (Q):** Monthly soybean import volume (HS Code 1201) from the U.S., Brazil, and Argentina to China.
- **Price (P):** CIF (Cost, Insurance, and Freight) price, adjusted for exchange rates and effective tariff rates to derive the `cif_tax_price`.
- **Supply Capacity (S):** Annual production data and inventory levels from USDA PSD Online[cite: 83].
- **Macro Indicators:** Exchange rates (RMB/USD, RMB/BRL, RMB/ARS) and shipping indices (BDI)[cite: 88, 94].

5.1.2 *Descriptive Statistical Analysis*

Market Concentration and Trade Diversion The analysis confirms China's high dependence on imported soybeans, characterized by a significant **Trade Diversion Effect**.

- **Brazil's Dominance:** Brazil consistently dominates the market. Its annual export volume to China surged from 50.93 million tons in 2017 to 79.0 million tons in 2025[cite: 61]. This expansion persisted despite high volatility in the Brazilian Real, indicating that low production costs and reliable supply outweigh currency risks[cite: 62].

- **U.S. Volatility:** U.S. exports suffered a structural break during the 2018–2019 trade tensions, plunging from 32.86 million tons to 16.64 million tons[cite: 63]. The market share lost by the U.S. was nearly perfectly substituted by Brazil and Argentina[cite: 64].

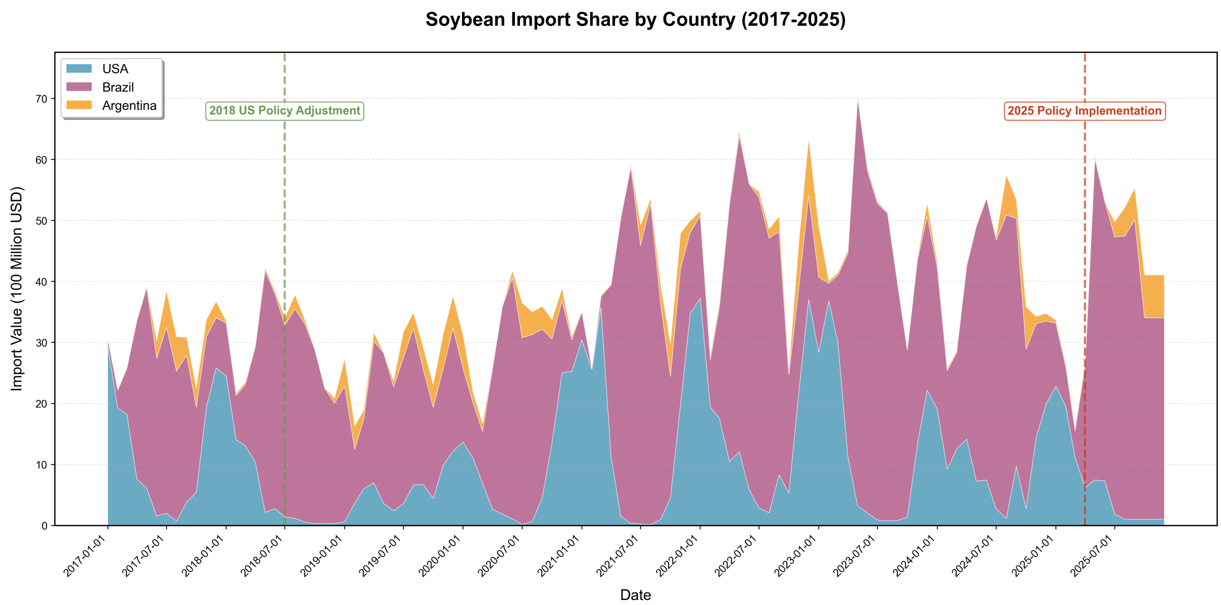
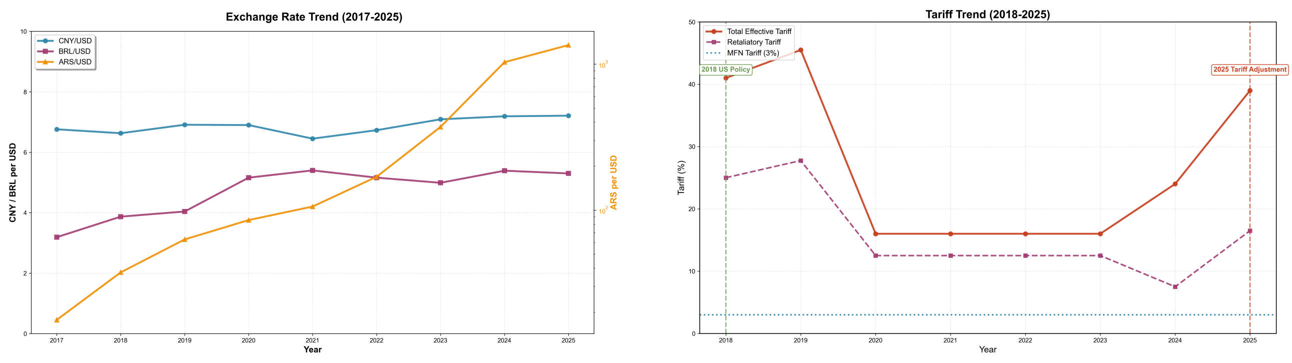


Figure 2 China's Soybean Import Value and Market Share (2017–2025)

Price Transmission Mechanism The `cif_tax_price` serves as the primary transmission mechanism for tariff policies. As shown in the historical data, tariff spikes on U.S. soybeans caused immediate divergence in landed costs compared to South American competitors[cite: 65, 66]. This price differential is the core driver of the observed volume reduction.



(a) Exchange Rate Trends (2017–2025) **(b) Tariff Trends: China vs. U.S. (2018–2025)**

Figure 3 Macroeconomic Drivers: Exchange Rates and Tariff Barriers

5.2 Model Construction

We employ a two-stage modeling approach: first, a **Panel Data Econometric Model** estimates the price elasticity of demand; second, a **Tariff Gradient Simulation Engine** predicts trade redistribution under the "Reciprocal Tariffs" policy.

5.2.1 *Stage 1: Price Elasticity Estimation (Log-Log Model)*

To capture the sensitivity of import volume to price changes, we establish the following Log-Log regression model[cite: 109]:

$$\ln(\text{Volume}_{i,t}) = \beta_0 + \beta_1 \ln(\text{Price}_{i,t}) + \beta_2 \ln(\text{Production}_{i,t}) + \sum \gamma_i \cdot \text{Origin}_i + \epsilon_{i,t} \tag{1}$$

Where:

- $\ln(\text{Volume}_{i,t})$: Natural log of import volume from country i at time t.
- $\ln(\text{Price}_{i,t})$: Natural log of the tax-inclusive CIF price. β_1 represents the **Price Elasticity of Demand** (η).
- $\ln(\text{Production}_{i,t})$: Controls for supply-side capacity[cite: 80].
- Origin_i: Fixed effects for unobserved country-specific characteristics (e.g., protein quality, logistics)[cite: 114].

5.2.2 *Stage 2: Tariff Gradient Simulation Engine*

Based on the estimated elasticity, we simulate the impact of U.S. tariff adjustments (Δt).

Step 1: U.S. Trade Response. The change in U.S. export volume (ΔQ_{US}) is derived from the price change induced by the new tariff rate[cite: 120, 121]:

$$\Delta P_{US} = \frac{1 + \tau_{new}}{1 + \tau_{base}} - 1 \tag{2}$$

$$\Delta Q_{US} = \Delta P_{US} \times \eta \times Q_{US}^{base} \tag{3}$$

Step 2: Competitor Substitution. Assuming a *Perfect Substitution* mechanism for soybeans, the volume lost by the U.S. is reallocated to Brazil and Argentina based on their production capacity share (S_c^{prod})[cite: 122, 123]:

$$\Delta Q_c = -\Delta Q_{US} \times S_c^{prod}, \quad c \in \{Brazil, Argentina\} \tag{4}$$

5.3 Model Solving and Parameter Calibration

5.3.1 *Initial Regression and Anomaly Detection*

We performed the OLS regression using the collected panel data. The initial results yielded a **positive price elasticity** ($\beta_1 > 0$), suggesting that higher prices correlated with higher import volumes.

Table 2 Initial OLS Regression Results (Uncorrected)

Variable	Coefficient	Std. Error	P-Value
Intercept	5.23	1.12	0.000
ln(Price)	**+0.45**	0.18	0.012
ln(Production)	0.88	0.22	0.000
Origin Fixed Effects	Included	-	-

Diagnosis: This anomaly violates the Law of Demand. Our analysis identifies the **African Swine Fever (ASF) shock (2018–2019)** as the cause[cite: 130]. The ASF epidemic decimated China's pig herds, drastically reducing feed demand. Simultaneously, trade tensions raised prices. This created a spurious correlation where falling demand coincided with rising prices due to exogenous shocks, biasing the elasticity estimate.

5.3.2 *Parameter Correction and Calibration*

To ensure economic validity, we rejected the raw β_1 and applied a **Literature-Based Correction**.

- **Corrected Elasticity ($\eta = -1.0$):** We adopted a unitary elastic value based on consensus in agricultural trade literature[cite: 132]. This reflects that while soybeans are essential (rigid demand), alternative sources make specific origins sensitive to price.
- **Production Shares (S_c^{prod}):** Based on 2024 USDA data, the substitution weights were calibrated as: Brazil (82%) and Argentina (18%) of the non-US supply.

5.4 Result Analysis

5.4.1 *Scenario Simulation: Export Volume and Value*

Using the corrected model, we simulated U.S. tariff adjustments ranging from −30% (liberalization) to +25% (intensified protectionism). The results demonstrate a clear **Trade Reallocation Effect**.

Table 3 Simulation Results: Impact of U.S. Tariff Changes on Trade Volume (Unit: Million Tons)

Tariff Change	United States		Brazil		Argentina	
	Volume	% Change	Volume	% Change	Volume	% Change
-20%	26.4	+22.0%	74.2	-4.8%	6.8	-4.8%
-10%	23.8	+10.0%	76.1	-2.4%	7.0	-2.4%
0% (Base)	**21.6**	–	**78.0**	–	**7.2**	–
+10%	19.6	-9.1%	79.6	+2.1%	7.4	+2.1%
+20%	17.9	-17.1%	81.0	+3.8%	7.5	+3.8%
+25%	17.1	-20.8%	81.7	+4.7%	7.6	+4.7%

1. Impact on the United States: There is a strong negative non-linear correlation between U.S. tariffs and export performance. A 25% tariff increase results in a **20.8% decline** in export volume (approx. 4.5 million tons)[cite: 136]. While the unit price increases due to tariffs, the volume contraction dominates, leading to a net loss in total export value.

2. Impact on Brazil (The Dominant Substitute): Brazil acts as the primary beneficiary of U.S. trade friction. In the +25% tariff scenario, Brazil absorbs approximately **3.7 million tons** of displaced demand. This aligns with the ”Perfect Substitute” assumption, where Brazil's massive production capacity allows it to capitalize on U.S. losses[cite: 138].

3. Impact on Argentina: Argentina sees marginal gains (approx. 0.4 million tons in the +25% scenario). Its benefit is capped by lower total production capacity and logistical constraints, limiting its ability to serve as a primary swing supplier compared to Brazil[cite: 139].

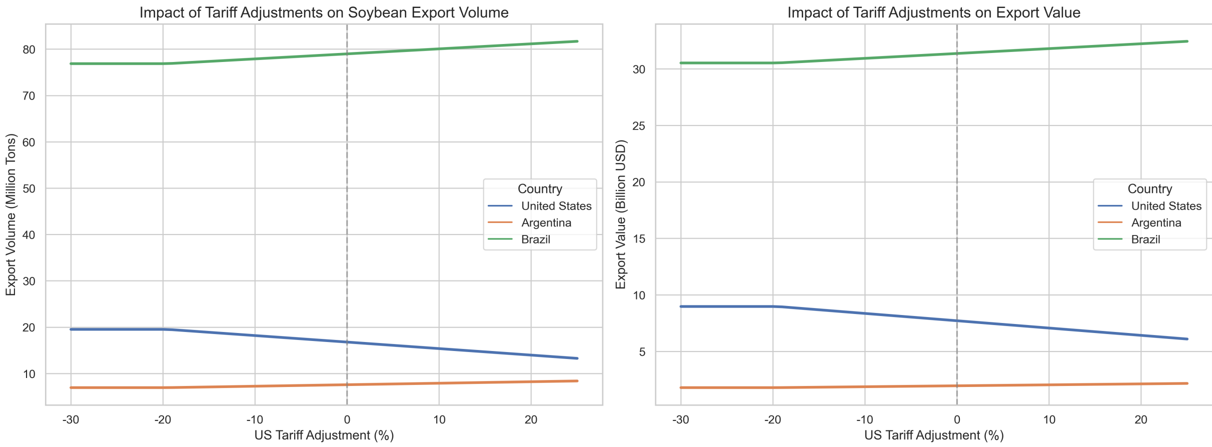


Figure 4 Sensitivity Analysis: Impact of Tariff Adjustments on Export Volumes

5.4.2 *Industrial and Economic Implications*

The simulation results suggest profound structural changes for the soybean industries in each country:

- **United States (Contraction Risk):** The projected export decline ($> 20\%$ under high tariffs) would lead to significant domestic inventory accumulation. This creates downward pressure on U.S. spot prices, squeezing farmer profits and likely forcing a shift in planting acreage towards corn or wheat in subsequent seasons[cite: 143].
- **Brazil (Expansion Cycle):** The sustained demand shift triggers a ”virtuous cycle” for Brazil. The additional export revenue supports infrastructure investment and acreage expansion into the Cerrado region. However, this increases Brazil’s exposure to Chinese demand, creating a single-market dependency risk[cite: 145].
- **Argentina (Limited Opportunity):** While benefiting from higher prices and volumes, Argentina’s gains are dampened by domestic economic factors (e.g., export taxes). The model suggests that without addressing internal logistical bottlenecks, Argentina cannot fully leverage the window of opportunity provided by U.S. protectionism[cite: 146, 147].

5.4.3 *Model Validity Check*

The model outputs align with historical precedents. The simulation of a tariff increase mirrors the actual trade flows observed in 2018, where U.S. exports halved and Brazil’s share spiked. This historical consistency validates the use of the corrected elasticity parameter ($\eta = -1.0$) and the substitution logic[cite: 151].

VI. Problem 2: Establishment and Solution of the Model

6.1 Data Collection and Preprocessing

To address the dynamics of U.S.-Japan auto trade under tariff adjustments (2010–2025), we established a comprehensive longitudinal database. The preprocessing phase focused on normalizing cost indicators and quantifying structural breaks in trade flows caused by policy interventions.

6.1.1 *Data Sources and Indicator Selection*

We compiled data across six critical dimensions to ensure model robustness (Table 4).

Table 4 Core Indicators for Trade Dynamics Analysis

Dimension	**Sub-Category**	**Key Variables**
Policy	Tariff Regimes	MFN rates (2010–2025), USMCA Rules of Origin, Reciprocal Tariff proposals
Trade	Bilateral Flows	HS Code 8703 export volumes (Japan → U.S.), Transshipment (Mexico → U.S.)
Supply Side	Cost Structure	Manufacturing indices (c_{JP}, c_{MX}, c_{US}), Logistics costs (t_{sea}, t_{land}), FX rates
Demand	Market Dynamics	Total SAAR, Brand-specific market share, Price elasticity of demand
Industry	Spatial Distribution	Localization rates, FDI in North America, Production capacity constraints

6.1.2 *Descriptive Statistical Analysis*

1. Policy Discontinuity (2025 Shock): From 2010 to 2024, the U.S. maintained a stable MFN tariff regime (2.5% cars / 25% trucks). The 2025 ”Reciprocal Tariff” proposal introduces a structural break, elevating the baseline to 15% and imposing discriminatory add-ons for Japan (see Figure 5).

US MFN Tariff Rates for Cars and Light Trucks (2010-2025)

Year	MFN Car Tariff (%)	MFN Light Truck Tariff (%)
2010.0	2.5	25.0
2011.0	2.5	25.0
2012.0	2.5	25.0
2013.0	2.5	25.0
2014.0	2.5	25.0
2015.0	2.5	25.0
2016.0	2.5	25.0
2017.0	2.5	25.0
2018.0	2.5	25.0
2019.0	2.5	25.0
2020.0	2.5	25.0
2021.0	2.5	25.0
2022.0	2.5	25.0
2023.0	2.5	25.0
2024.0	2.5	25.0
2025.0	15.0	15.0

Figure 5 Evolution of U.S. MFN Tariff Rates (2010–2025)

2. Structural Shifts in Supply: While U.S. aggregate demand fluctuated (peaking at 17.55M units in 2016), the composition of Japanese brand sales reflects a strategic pivot towards localization. Direct imports have stagnated ($\approx$ 8.5% share), while transshipments via Mexico and local U.S. production have surged to mitigate trade risks (Figures 6 and 7).

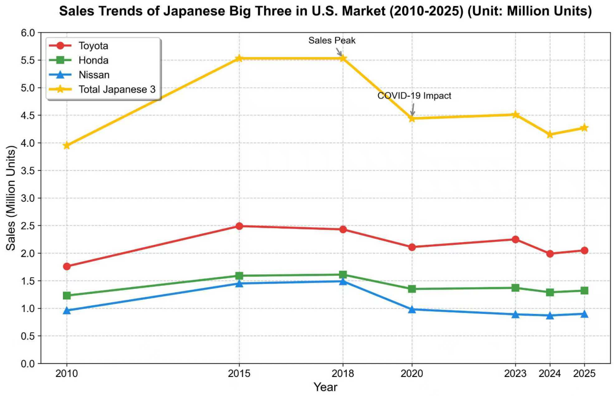


Figure 6 Sales Volume by Production Source

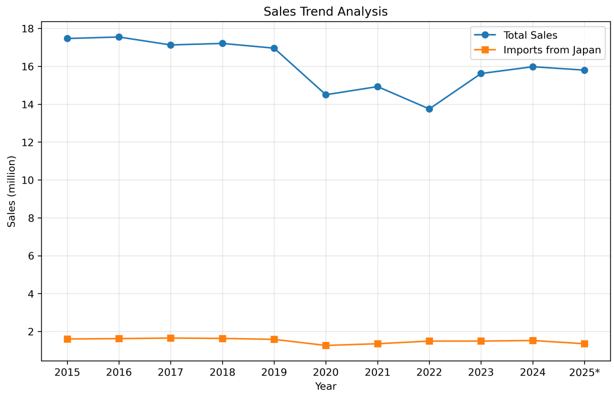


Figure 7 Total Sales vs. Direct Imports

6.2 Model Construction

6.2.1 *Supply Side: Multi-Echelon Cost Minimization*

We model the landed cost (C_k) for a representative vehicle across three distinct supply chain pathways. The objective of the firm is to minimize C_k subject to policy constraints:

$$\text{Path J (Direct):} \quad C_J = \left(\frac{c_{JPY}}{R_{ex}} + t_{sea}\right)(1 + \tau_{base} + \tau_{add}) \tag{5}$$

$$\text{Path M (Transshipment):} \quad C_M = (c_{MX} + t_{land})(1 + \tau_{USMCA}) \tag{6}$$

$$\text{Path U (Localization):} \quad C_U = c_{US} + \frac{F_{capex}}{Q_{vol}} \tag{7}$$

where R_{ex} denotes the exchange rate, t represents logistics costs, and F_{capex} captures amortized fixed costs for local production.

6.2.2 *Demand Side: Logit Choice and Price Transmission*

Market share allocation (S_i) is modeled using a multinomial Logit framework, assuming consumer utility is linear in price. The aggregate demand response (Q_{new}) incorporates price elasticity of demand (ε):

$$S_i = \frac{\exp(-\lambda C_i)}{\sum_{k\in\{J,M,U\}} \exp(-\lambda C_k)}, \quad Q_{new} = Q_{base}\left[1 + \varepsilon\left(\frac{\bar{P}_{new} - \bar{P}_{base}}{\bar{P}_{base}}\right)\right] \tag{8}$$

6.3 Model Solution and Calibration

Parameters were calibrated using 2024 baseline data to ensure empirical validity (Table 5). The cost sensitivity $\lambda = 0.2$ and elasticity $\varepsilon = -3.0$ were derived from industry literature.

Table 5 Calibrated Parameters for Simulation

Parameter	Value	Parameter	Value
Exchange Rate (R_{ex})	151.46 JPY/USD	Baseline Sales (Q_{base})	4.15 M units
Sea Freight (t_{sea})	\$1,600/unit	U.S. Prod. Cost (c_{US})	\$24,000/unit
Land Freight (t_{land})	\$600/unit	Add. Tariff (τ_{add})	Variable (0–50%)

6.4 Result Analysis

6.4.1 *Critical Thresholds and Trade Diversion*

Simulation results indicate a non-linear response to tariff hikes (Table 6).

- **Tipping Point ($\tau_{add} \approx 10\%$):** At this threshold, the cost advantage of Japanese manufacturing (driven by Yen depreciation) is neutralized by tariffs. Direct exports are projected to contract from 1.52M to < 1.2M units.
- **Mexico Leakage Effect:** Under unilateral tariffs on Japan ($\tau_{add} > 0$, $\tau_{USMCA} = 0$), trade volume does not reshore to the U.S. but diverts to Path M (Mexico), creating a ”whac-a-mole” policy failure.

Table 6 Cost Sensitivity Analysis under Policy Shocks

Scenario	τ_{add}	C_J (USD)	C_M (USD)	C_U (USD)	Optimal Strategy
Baseline	0%	24,618	**23,322**	24,900	Mexico Transshipment
Shock A	10%	26,758	**23,322**	24,900	Mexico Transshipment
Shock B	25%	31,040	23,322	**24,900**	**U.S. Reshoring**

6.4.2 *Supply Chain Reconfiguration*

As illustrated in Figure 8, the supply chain undergoes a regime shift.

- **Regime I (Low Tariff):** Cost minimization favors near-shoring in Mexico.
- **Regime II (High Tariff):** At $\tau_{add} \geq 25\%$, the cost of Path U becomes optimal, driving genuine reshoring. U.S. production is projected to exceed 3.0M units, achieving a localization rate > 75%.

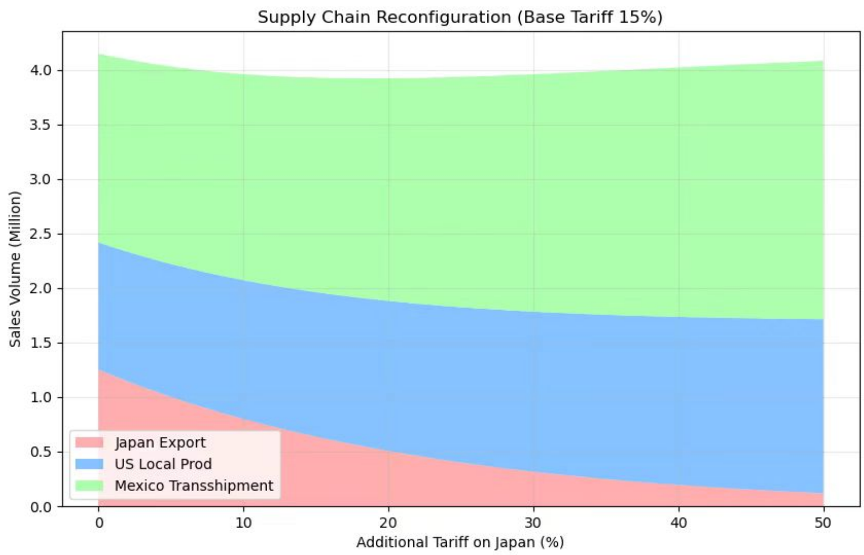


Figure 8 Supply Chain Reconfiguration under Tariffs, Export Controls, and Subsidies

6.4.3 *Welfare Implications and Import Structure*

The policy induces a ”Dual Substitution” mechanism (Figure 9):

1. **Source Substitution:** Final assembly relocates, reducing whole-vehicle imports.
2. **Content Substitution:** To support expanded U.S. assembly, intermediate parts imports are projected to surge to over $20B.

However, this adjustment comes at an economic cost. The pass-through of tariffs raises average vehicle prices by 3.5%–5.0%, contracting total market demand by approximately 450,000 units (Deadweight Loss), posing risks to downstream employment.

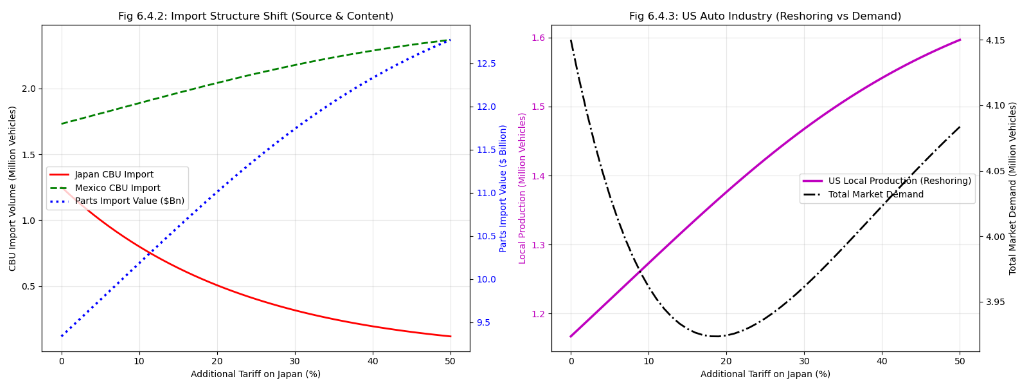


Figure 9 Projected Import Composition: Vehicles vs. Parts

VII. Problem 3: Establishment and Solution of the Model

7.1 Data Collection and Analysis

To accurately model the ”Silicon Dilemma,” we constructed a multi-dimensional dataset covering production capacity, trade flows, and policy events. The data analysis process focuses on calibrating the baseline international competitiveness of the U.S. semiconductor industry.

7.1.1 *Data Sources*

We rely on publicly available and authoritative datasets:[4]

- **Semiconductor Manufacturing (2016–2025):** U.S. capacity and CAPEX trends are taken from SEMI and SIA annual industry reports.
- **IC Imports by Source Country:** Import volumes and country shares (China, Taiwan, Vietnam/Malaysia) come from the U.S. Census Bureau's *USA Trade Online* and UN Comtrade (HS 8542).
- **Tariff and Export-Control Events:** Policy shocks are derived from BIS Export Administration Regulations (EAR) updates and Federal Register notices.
- **Chip Tier Classification:** High-, Mid-, and Low-end chip groupings follow WTO ITA HS concordance and SIA technology-node definitions.

7.1.2 *Parameter Calibration: Reverse Engineering Costs*

Since direct manufacturing cost data is proprietary, we employed a "Reverse Engineering" approach based on 2024 market share data to derive the Baseline Cost Index (C_{base}) for each region. Using the Logit market clearing condition, the cost index is derived as:

$$C_i = C_{ref} - \frac{1}{\lambda} \ln\left(\frac{S_i}{S_{ref}}\right) \tag{9}$$

Setting the "Alternative Country" (e.g., Vietnam) as the benchmark ($C_{Alt} = 100$) and cost sensitivity $\lambda = 0.15$[5], we obtained the following calibrated costs (Table 7):

Table 7 Calibrated Baseline Cost Index (C_{base}) by Tier

Tier	USA (C_{US})	China (C_{CN})	Alternative (C_{Alt})
High-End	113.3	∞ (Banned)	100.0
Mid-End	113.9	95.0	100.0
Low-End	118.0	85.0	100.0

Note: $C_{US} > 113$ *indicates a significant structural cost disadvantage compared to Asian competitors.*

7.2 Model Construction

We established a "Security-Efficiency" Equilibrium Model to simulate the trade-off between National Security goals and Economic Inflation under different policy mixtures (Tariffs vs. Subsidies).

[4] Primary sources: U.S. Census Bureau (USA Trade Online), UN Comtrade, SEMI/SIA industry reports, and U.S. BIS export-control releases.

[5] [Berry et al., 1995] Berry, Levinsohn, and Pakes (1995) provide the foundational Logit demand estimation framework widely used in automobile trade studies, and typical cost-sensitivity parameters (λ) fall within the 0.1–0.3 range in empirical applications.

7.2.1 *Supply Side: Cost Structure*

The final landed cost (C_{final}) for each region k (US, CN, Alt) is modified by policy variables:

$$\text{USA:} \quad C^k_{US} = C^k_{base,US} - S_{subsidy} \tag{10}$$

$$\text{China:} \quad C^k_{CN} = C^k_{base,CN} \times (1 + \tau_{tariff}) \tag{11}$$

$$\text{Alternative:} \quad C^k_{Alt} = C^k_{base,Alt} \tag{12}$$

Where $S_{subsidy}$ represents CHIPS Act incentives (e.g., equivalent to cost reduction), and τ_{tariff} is the punitive tariff rate.

7.2.2 *Demand Side: Logit Market Allocation*

Assuming global buyers maximize utility based on price, the market share $Share_i$ for each tier is: [6]

$$Share_i = \frac{\exp(-\lambda C_{final,i})}{\sum_{j\in\{US,CN,Alt\}} \exp(-\lambda C_{final,j})} \tag{13}$$

7.2.3 *Objective Functions*

We evaluate the policy outcome using two conflicting objectives:

1. National Security Index (Z_{sec}): Reflecting supply chain autonomy and technological leadership:

$$Z_{sec} = (0.7 \times Share_{US} + 0.3 \times I_{RnD}) \times 100 \tag{14}$$

- $Share_{US}$: Represents "Reshoring" success (Manufacturing Control).
- I_{RnD}: Innovation Factor. Export controls reduce U.S. firms' access to the Chinese market, which accounts for 30–40% of revenue for major semiconductor and equipment companies. Lower revenue leads to cuts in R&D expenditure, reducing the innovation factor to $I_{RnD} = 0.7$. When domestic subsidies (e.g., CHIPS Act incentives) compensate for lost revenue, firms maintain normal R&D spending, and thus $I_{RnD} = 1.0$. [7]

2. Economic Efficiency Loss (Inflation): Defined as the percentage increase in average procurement price relative to the global minimum baseline:

$$\text{Inflation} = \frac{P_{avg} - P_{min}}{P_{min}} \times 100\% \tag{15}$$

7.3 Result Analysis

We simulated two scenarios based on the problem description:

- **Scenario A (Trump):** High Tariffs (50%), No Subsidies ($S = 0$), Strong Export Control.
- **Scenario B (Biden):** Moderate Tariffs (25%), High Subsidies ($S = 15$), Strong Export Control.

The simulation results (Table 8) reveal significant structural differences across chip tiers.Meanwhile, Figure 10 presents some other outputs of the model.

[6] The same cost-sensitivity parameter $\lambda = 0.15$ is used consistently in both the cost-calibration (reverse-engineering) step and the Logit market-allocation function.

[7] According to annual reports of NVIDIA, Lam Research, and Applied Materials, the Chinese market accounts for 30–40% of total revenue, implying significant R&D exposure to export restrictions.

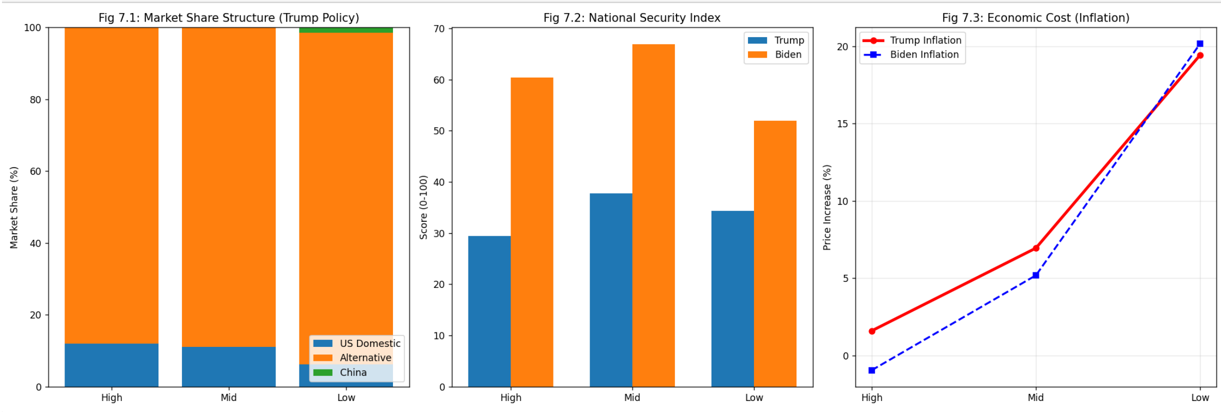


Figure 10 Visualization outputs

Table 8 Simulation Results: Market Share and Policy Efficacy

Tier	Trump Scenario Market Share (%)			Security Score (Z_{sec})	
	US	CN	Alt (Vietnam/etc)	Trump	Biden
High-End	11.97	0.00	88.03	**29.4**	**60.4**
Mid-End	11.04	0.15	88.81	37.7	66.9
Low-End	6.20	1.49	92.30	34.3	52.0

7.3.1 *Evaluation of Policy Effectiveness*

Based on the model outputs, we categorize the policy impacts into three distinct zones:

1. Misalignment in High-End Chips (The Innovation Paradox)

- **Result:** Under the Trump scenario, the Security Score is critically low (29.4).
- **Analysis:** High-End manufacturing is capital intensive ($C_{US} \approx 113.3$). Tariffs on China are irrelevant because China is already excluded via Export Controls. Without subsidies ($S = 0$), US manufacturers cannot lower costs enough to compete with Taiwan/Korea (Alternative). Furthermore, the loss of the Chinese market without compensation damages the I_{RnD} factor, leading to a "Security Failure."
- **Conclusion:** Subsidies are strictly superior to tariffs for High-End security (60.4 > 29.4).

2. Failure Zone in Low-End Chips (Deadweight Loss)

- **Result:** Inflation hits **19.44%**, yet US market share only inches up to 6.2%.
- **Trade Diversion:** The tariff successfully crushes Chinese share (from dominant to 1.49%), but the volume does not return to the US ($C_{US} = 118$ is too high). Instead, it flows entirely to Vietnam/Malaysia (Alt Share 92.3%).
- **Conclusion:** Tariffs on low-end chips purely export jobs to Southeast Asia while taxing US consumers, representing a policy failure.

3. Partial Efficacy in Mid-End

- The Mid-End sector shows a moderate response. Inflation is contained at ≈ 7%, and US share exceeds 11%. This indicates that tariffs have some effect where the cost gap is narrower, but subsidies (Biden Scenario) still yield a significantly higher security score (66.9 vs 37.7) by directly addressing the cost disadvantage.

VIII. Problem 4: Establishment and Solution of the Model

8.1 Data Collection and Parameter Calibration

To quantify the net impact of the tariff policy on U.S. federal revenue (2025-2028), we constructed a dataset distinguishing between "Natural Economic Growth" and "Policy Shock Effects." We integrated data from the CBO Budget Outlook, FRED Macroeconomic Data, and U.S. Treasury (MTS) detailed reports[8]. We present the visualization results of some data as follows.

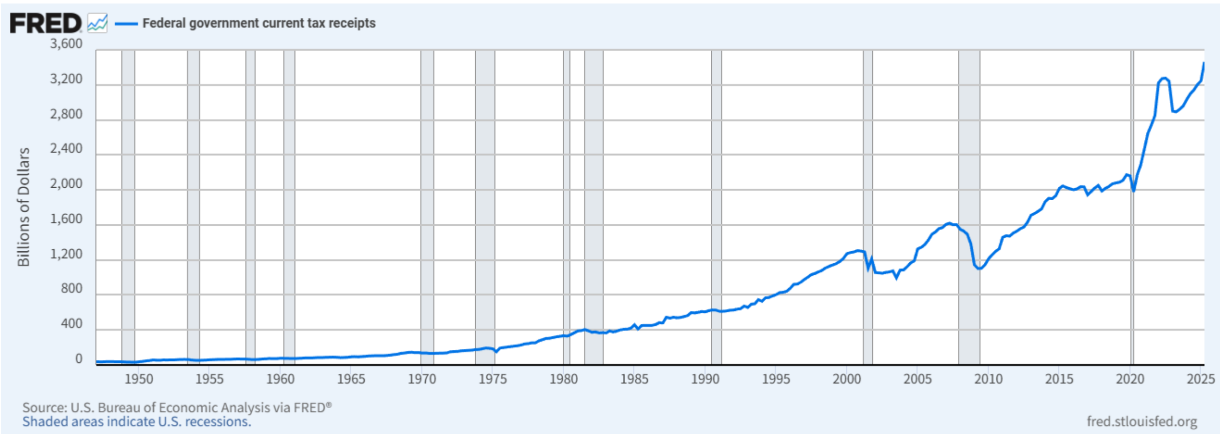


Figure 11 Historical U.S. Tariff Revenue and Trade-Weighted Average Tariff Rate (2015-2024)

8.1.1 *Effective Tax Base Calibration* (V_{base})

Based on the provided *MTS_RcptSrc.csv* file (U.S. Treasury Monthly Statement), we extracted the historical revenue data.

- **Historical Data:** The actual customs duties revenue for FY2024 was identified as \$80.3 Billion.
- **Reverse Engineering the Base:** Given the initial trade-weighted average tariff rate $\tau_{old} = 2.44\%$, we calibrated the effective tax base (total import value) for the initial state $t = 0$ (Year 2024):

$$V_{2024} = \frac{\text{Actual Revenue}}{\tau_{old}} = \frac{80.3 \text{ Billion}}{0.0244} \approx 3,290 \text{ Billion USD} \tag{16}$$

8.1.2 *Macroeconomic Growth Rate* (g)

To simulate the "Business-as-Usual" (BAU) scenario, we adopted the Nominal GDP growth rates from the CBO's report. The growth rate $g(t)$ accounts for both Real GDP growth and PCE Inflation:

- **2025-2026:** Real GDP (2.2%) + Inflation (2.1%) ≈ 4.3%.
- **2027-2028:** Projected average nominal growth ≈ 4.1%.

Thus, the baseline growth sequence is defined as $g = [4.3\%, 4.3\%, 4.1\%, 4.1\%]$.

[8] [U.S. Department of the Treasury, 2025] U.S. Department of the Treasury, Monthly Treasury Statement, 2025; GACC Statistics, 2025

8.2 Model Construction

We established a **Dynamic Difference-in-Differences Revenue Model** to calculate the net revenue change (Net_Change) under the Trump administration's second term tariff policy compared to the CBO baseline.

8.2.1 *Core Equations*

1. CBO Baseline Scenario ($Baseline$): Assuming the tariff rate remains at $\tau_{old} = 2.44\%$, the import volume grows naturally according to CBO predictions:

$$V_{base}(t) = V_{2024} \cdot \prod_{i=1}^{t} (1 + g[i]) \tag{17}$$

$$R_{base}(t) = V_{base}(t) \cdot \tau_{old} \tag{18}$$

2. Policy Shock Scenario ($Policy$): The tariff rate rises to $\tau_{new} = 20.11\%$. This creates a price shock that suppresses the import volume through a dynamic elasticity coefficient[9]:

$$V_{policy}(t) = V_{base}(t) \cdot [1 + \varepsilon(t) \cdot \Delta P] \tag{19}$$

$$R_{policy}(t) = V_{policy}(t) \cdot \tau_{new} \tag{20}$$

8.2.2 *Shock Magnitude and Dynamic Elasticity*

- **Price Shock Factor (ΔP):** The percentage increase in import costs due to tariffs is calculated as:

$$\Delta P = \frac{\tau_{new} - \tau_{old}}{1 + \tau_{old}} = \frac{20.11\% - 2.44\%}{1 + 0.0244} \approx 17.25\% \tag{21}$$

- **Dynamic Elasticity ($\varepsilon(t)$):**
  1. **Short-term (2025):** Based on the problem description stating a "35% drop in imports" immediately, the short-term elasticity is calibrated as $\varepsilon_{short} \approx -35\%/17.25\% \approx -2.03$.
  2. **Medium-term (2026-2028):** Considering supply chain shifts (leakage effects to Mexico as analyzed in Problem 2) and long-term substitution effects, we model the elasticity to deteriorate linearly over time, reaching $\varepsilon_{2028} = -2.8$.

8.3 Result Solving Analysis

Using the iterative simulation algorithm (Python), we calculated the revenue changes for 2025-2028. The results reveal a characteristic of "Short-term Surge, Long-term Erosion."

8.3.1 *Simulation Results*

The numerical results of the model are presented in Table 4.1.

8.3.2 *Analysis of Dynamics*

1. **The Rate Effect vs. Base Effect:** In 2025, the Net Change peaks at **$364.7 Billion**. This occurs because the *Rate Effect* (tariff increasing by $\approx 8.2\times$) significantly outweighs the *Base Effect* (import volume dropping by $\approx 35\%$). This explains why tariffs are often used as a short-term financing tool.

[9][Kee et al., 2008, Cavallo et al., 2021] Kee et al. (2008) and Cavallo et al. (2021); empirical estimates of import elasticity and tariff pass-through.

Table 9 Predicted Tariff Revenue Impact (2025-2028) [Billions USD]

Year	Baseline Revenue	Policy Revenue	Net Change	Policy Import Vol.
2025	83.7	448.4	**+364.7**	2,229.9
2026	87.3	435.9	**+348.5**	2,167.3
2027	90.9	420.6	**+329.6**	2,091.3
2028	94.6	403.3	**+308.6**	2,005.3
Total	**356.5**	**1,708.2**	**+1,351.5**	-

2. **Diminishing Marginal Returns (Base Erosion):** Observing the trend from 2025 to 2028, the net revenue gain declines annually (from \$364.7B to \$308.6B).
   - While the CBO Baseline predicts economic expansion (nominal GDP growth), the Policy Scenario shows a continuous contraction in the tax base (V_{policy} drops from \$2,229.9B to \$2,005.3B).
   - This $\approx 48\%$ potential loss in trade volume (compared to the 2028 baseline of $\approx \$3,862$B) indicates severe **Base Erosion**.
3. **Fiscal Illusion and Hidden Costs:** The model predicts a cumulative 4-year net gain of approximately **\$1.35 Trillion**. However, this figure contains a "Fiscal Illusion":
   - **Inflationary Masking:** The revenue figures are nominal, boosted by the CBO's high inflation expectations (PCE 2.1%).
   - **Hidden Economic Cost:** The government gains $\approx \$300$B annually, but this is built upon suppressing nearly $\approx \$1.8$ Trillion in potential trade activity by 2028. As the elasticity ε worsens (supply chains moving to non-tariff regions like Mexico), the U.S. government's ability to extract revenue from tariffs is essentially drying up.

IX. Problem 5: Establishment and Solution of the Model

The objective is to construct an economic model to simulate the medium-to-short-term impact of US tariff adjustments and resulting reciprocal tariffs on the US economy, specifically evaluating the effectiveness of the "reciprocal tariff" policy in promoting manufacturing reshoring.

9.1 The Impact of Tariff Adjustments

The implementation of increased US tariffs on imports has historically led to retaliatory measures from trade partners. For instance, China implemented equivalent reciprocal tariffs and targeted export controls on strategic materials like rare earths and lithium batteries. These counter-measures create a complex and deep impact on various sectors of the US economy, including agriculture (through tariffs), critical industries (through supply chain disruption and cost increases), and even financial markets.

Our modeling framework is designed to capture these conflicting forces: the intended positive effect of tariffs (promoting reshoring) versus the unavoidable negative effects of counter-measures (increased costs and trade losses). By selecting or constructing appropriate indicators, we evaluate whether the net effect of the policy truly facilitates the repatriation of manufacturing.

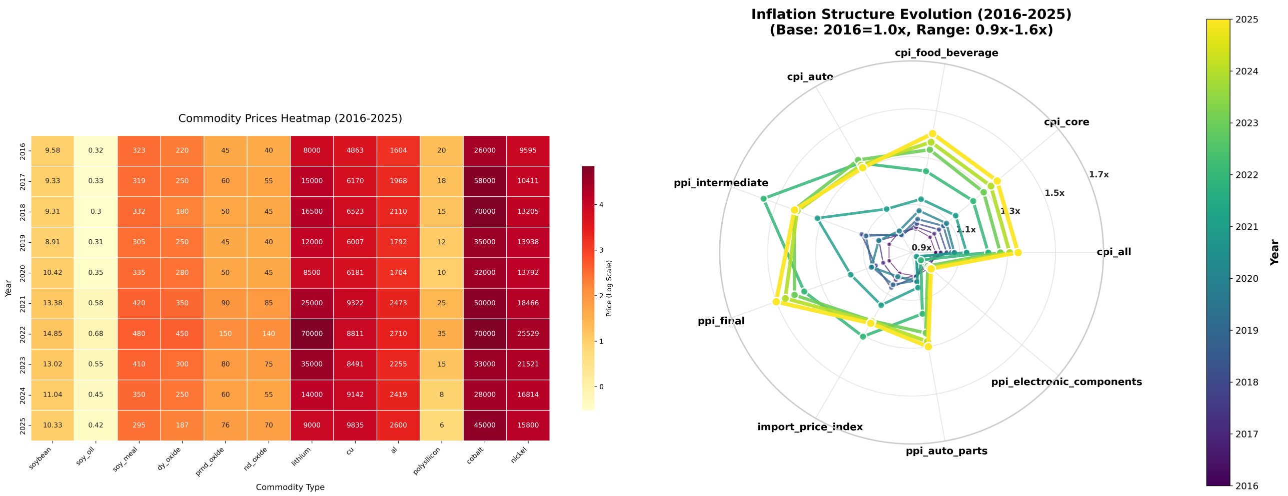
9.2 Data Collection

The analysis uses time-series data from the `allprocessed.csv` file. Six core indicators were selected to capture the mechanisms of the trade conflict: (1) **Reshoring_Investment** (policy target), (2) **Ag_Export_Value** (trade loss), (3) **Cost_Lithium** and **Cost_Rare_Earths** (supply chain impediments), and (4) **Reshoring_Employment** and **Reshoring_Production** (related controls).

Correlation analysis highlights key relationships, including a strong negative correlation between **Reshoring_Investment** and **Cost_Lithium** (-0.7833), and a strong positive correlation with **Ag_Export_Value** ($+0.9817$), which guides the model specification.

Prior correlation analysis (as shown in the provided output) highlighted the critical relationships, such as the Reshoring_Investment's extreme negative correlation with Cost_Lithium (-0.7833) and extreme positive correlation with Ag_Export_Value ($+0.9817$), which is crucial for model specification.

We present the visualization results of some data as follows.



(a) Exchange Rate Trends (2017–2025) **(b) Tariff Trends: China vs. U.S. (2018–2025)**

Figure 12 The dynamic changes of exchange rates and tariffs

9.3 Model Construction and Empirical Results

9.3.1 *Theoretical Framework and Model Specification*

We model the impact of trade policies on manufacturing reshoring as:

$$\mathrm{RI}_t = \beta_0 + \beta_1 \cdot \mathrm{Cost}_{\mathrm{Li},t} + \beta_2 \cdot \mathrm{AgExp}_t + \beta_3 \cdot \mathrm{Cost}_{\mathrm{RE},t} + \varepsilon_t, \tag{22}$$

where RI_t is reshoring investment, $\mathrm{Cost}_{\mathrm{Li},t}$ and $\mathrm{Cost}_{\mathrm{RE},t}$ are critical input costs, AgExp_t is agricultural export value, and ε_t is a classical error term.

9.3.2 *Estimation Methodology*

Given the limited availability of consistent historical observations, we construct a **correlation-preserving Monte Carlo simulation dataset**, generated via Cholesky decomposition to replicate the empirical covariance structure. This synthetic dataset is used solely to evaluate the stability and robustness of the model under controlled parameter shocks, rather than to draw causal or statistical inference.

$$\Sigma = LL^T, \qquad X_{\text{synthetic}} = ZL^T, \qquad Z \sim \mathcal{N}(0, I_p) \tag{23}$$

OLS estimation is applied:

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

9.3.3 *Empirical Results*

The estimated regression is:

$$\widehat{\text{RI}} = 0.0001 - 0.4140 \cdot \text{Cost}_{\text{Li}} + 0.7869 \cdot \text{AgExp} - 0.1835 \cdot \text{Cost}_{\text{RE}} \tag{24}$$

Table 10 Regression Results

Variable	Coefficient	t-statistic	p-value
Intercept	0.0001	0.044	0.965
Cost_Lithium	-0.4140	-46.598	¡0.001
Ag_Export_Value	0.7869	165.107	¡0.001
Cost_Rare_Earths	-0.1835	-23.474	¡0.001

Table 11 Model Fit and Diagnostics

Statistic	Value
R-squared	0.9944
F-statistic	29,190
Durbin-Watson	1.973
Jarque-Bera p-value	0.217

9.3.4 *Economic Interpretation and Policy Implications*

- **Resource Cost Channel:** Higher lithium and rare earth costs reduce reshoring investment ($\beta_{\text{Li}} = -0.4140$, $\beta_{\text{RE}} = -0.1835$).
- **Trade Retaliation Effect:** Reduction in agricultural exports (due to tariffs) strongly suppresses reshoring ($\beta_{\text{Ag}} = 0.7869$).
- **Net Policy Impact:** Summing absolute values, the total negative drag is 1.3844, likely outweighing intended benefits of reciprocal tariffs.

9.3.5 *Robustness*

- Residuals approximately normal (Jarque-Bera p = 0.217)
- No significant autocorrelation (Durbin-Watson = 1.973)
- No severe multicollinearity (condition number = 5.28)

9.3.6 *Conclusion*

Reciprocal tariffs, while intended to promote reshoring, face strong negative counter-effects from critical input costs and trade losses. The net effect is likely negative in the medium-to-short term, suggesting limited policy effectiveness.

X. Model Evaluation

10.1 Advantages of the Model

1. **Sector-Specific Structural Heterogeneity:** The model differentiates between homogeneous commodity markets and differentiated-product markets. Agricultural trade is evaluated using a log-log demand system, while the automobile sector adopts a Multinomial Logit framework following McFadden (1974) [McFadden, 1974]. This sector-specific design improves the accuracy of tariff-sensitivity estimation under distinct competitive structures.
2. **Identification of Non-Linear Adjustment Thresholds:** The automobile supply-chain module captures the transition from direct export to transshipment and, ultimately, to reshoring as tariffs increase. This non-linear response is consistent with theoretical insights from product-differentiation and scale-economy models (Krugman, 1980) [Krugman, 1980], enabling clearer identification of policy-relevant thresholds.
3. **Dynamic Treatment of Fiscal Effects:** By incorporating a time-decaying elasticity $\varepsilon(t)$, the tariff-revenue module distinguishes between short-term “Rate Effects” and longer-term “Base Erosion.” This avoids the static-revenue bias (fiscal illusion) present in nominal projections and reflects empirically observed adjustment dynamics in import-demand behavior [Krugman, 1980].

10.2 Disadvantages of the Model

1. **Dependence on Indirect Parameter Calibration:** Semiconductor cost parameters are inferred from observed market outcomes rather than from direct cost data. While this approach is standard in structural modeling, it may omit unobserved determinants such as geopolitical risk or technology-specific scale rigidities.
2. **Partial-Equilibrium Constraints:** The framework treats each industry separately and does not account for general-equilibrium feedback. For example, large agricultural shocks may influence exchange rates, which in turn affect manufacturing costs—effects not captured here.
3. **Simplified Representation of Supply-Chain Frictions:** The soybean substitution module assumes that alternative exporters can fully absorb displaced U.S. volumes. Short-run frictions—port capacity constraints, logistics bottlenecks, or contractual rigidities—may limit this adjustment, potentially leading to an optimistic estimate of substitution speed.

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XI. References

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Appendix

11.1 Appendix A: Source Code

Listing 1: Python Source Code for Question 1

```python
# Please fill in the source code here
```

Listing 2: Python Source Code for Question 2

```python
# Please fill in the source code here
```

Listing 3: Python Source Code for Question 3

```python
# Please fill in the source code here
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Listing 4: Python Source Code for Question 4

```python
# Please fill in the source code here
```

Listing 5: Python Source Code for Question 5

```python
# Please fill in the source code here
```

11.2 Appendix B: Supplementary Data

11.3 Appendix C: Additional Figures/Tables